DESIGN AND MANUFACTURE OF MOBILE ACTUATOR FOR ANKLE EXOSKELETON

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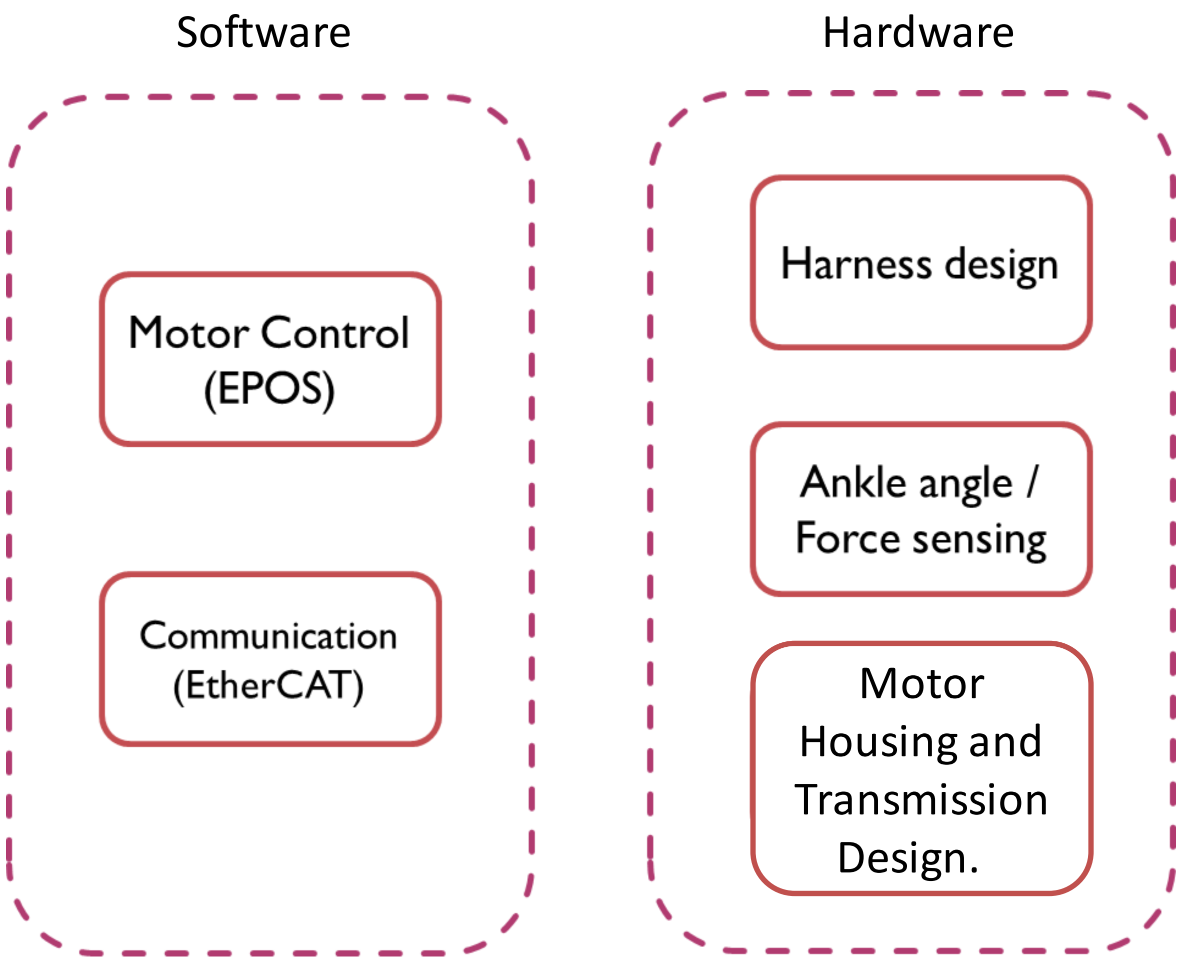
1. PROJECT OVERVIEW

Objectives:

- Develop a suit
  - Deliver the motor assistance to leg
  - Design and Manufacturing the Motor housing
  - Design and Manufacturing the Transmission

- Following Torque command accurately
  - Implement EtherCAT
  - Analog Control
  - Obtain the performance of motor



2.1 MOTIVATION

- Over the last two decades, several lower extremity robotic exoskeleton systems have been developed for tasks ranging from heavy lifting to helping the paralyzed, disabled or elderly to walk.
- This research aims to design and manufacture the exo suit and implement EtherCAT protocol for the ankle exoskeleton which will reduce the lag time of the torque command.
- Key motivation factors for all these wearable exoskeletons are minimizing the weight and making it more comfortable.

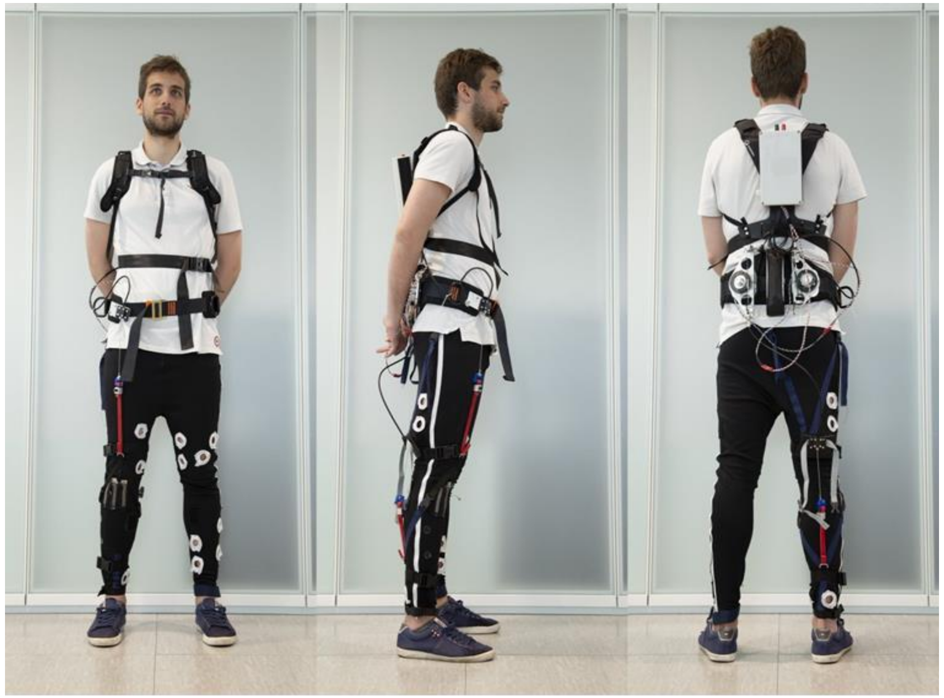


Fig1: XoSoft exo

2.2 CHALLENGES

The main challenges in this research are:

- Implementing EtherCAT protocol to the controller.

- Reducing the lag time from motor command to actuation of motor.

- Designing a lightweight exosuits that is comfortable to wear.
    - Motor Housing and Transmission mounted on the body.
    - Weight Distribution and force distribution are critical.
    - Increase comfort for long duration use.

Fig2: EtherCAT connection

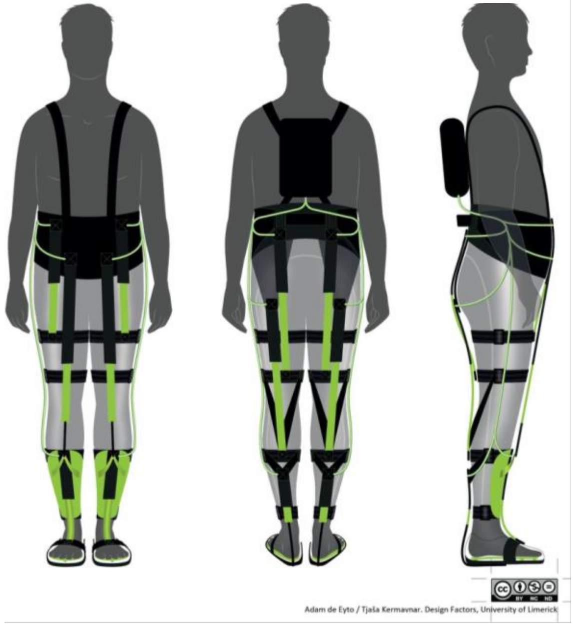
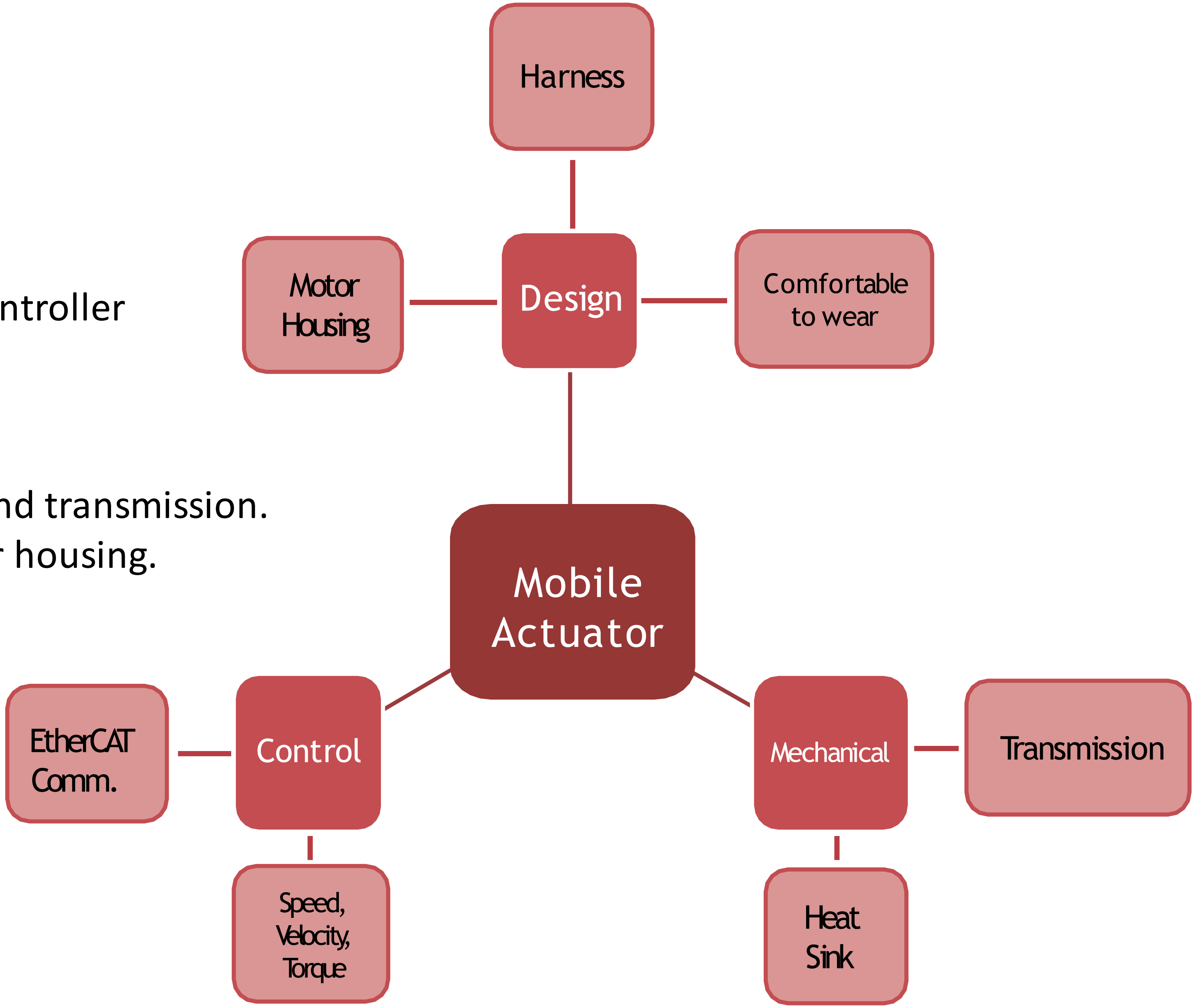


Fig3: Design Factors

2.3 OVERVIEW OF PROPOSED WORK

- Software Goals
  - Analog Control of motor using EPOS2 P Controller
  - Obtain the performance of motor

- Hardware Goals
  - Design and Manufacture Motor housing and transmission.
  - Design the harness mounting of the motor housing.
  - Design the Heat Sink.



2.4 INTELLECTUAL MERIT

- Implementation of EtherCAT in our research
    - Delay time reduces to 6-8ms in EtherCAT connection.

- Our Novel design of Motor housing and Transmission
    - Reduces the weight and bulkiness of the exosuit using light weight materials like Thermoplastic Polymer (TPE) and Aluminum.

- A Novel idea in the design of the transmission is "Adaptability"
    - Tension the pulley belts by adjusting the center to center distance.
    - Adjustability of the outer case makes it easier to increase or decrease gear ratios as required.

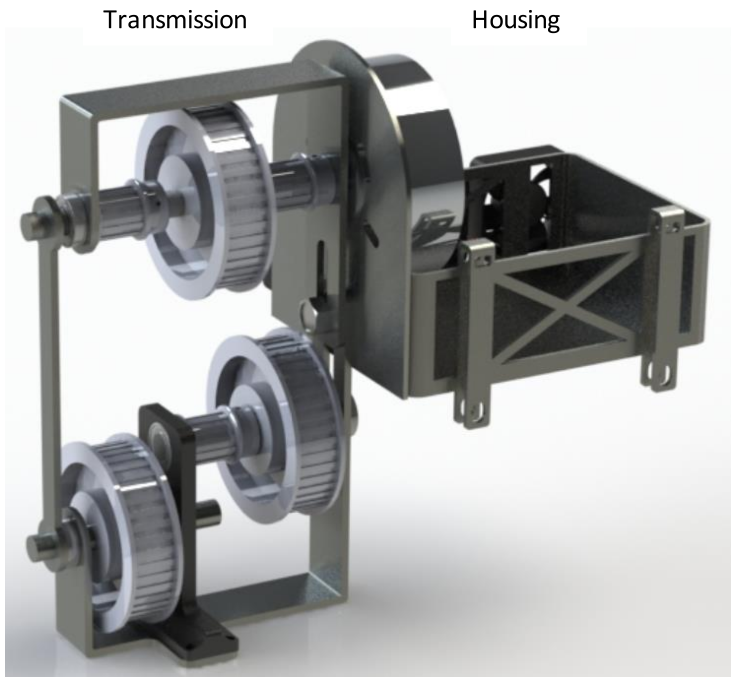


Fig4: CAD Model of transmission and Housing

2.5 BROADER IMPACTS

- Our lightweight design of the motor housing and transmission makes it more comfortable to wear.

- Our use of polymer materials also make the housing and transmission economical to manufacture making it more accessible for researchers to conduct research in this field.

- A cheaper and lighter housing and transmission enables it to be used in other applications like military operations to augment the abilities of the soldier and reduce fatigue of workers in their work daily activities.

Fig6: Soft exosuit to augment abilities.

Fig6: Harvard's Wyss Institute and Harvard SEAS soft exosuit

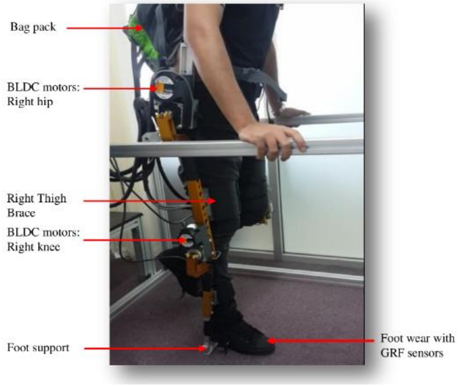
3.1 LITERATURE SURVEY, STATE OF ART

"Exoskeleton robot control for synchronous walking assistance in repetitive manual handling works based on dual unscented Kalman filter"

- DOF: 4 at each leg.
- Bi-directional brushless DC motor -Torque 38 Nm.

Drawback : Rigid links resist the movement of the biological joints.

Bulky and has lot of inertia which will increase the metabolic rate.

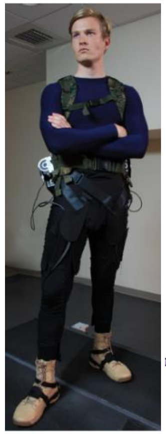


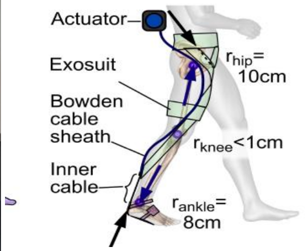
"Autonomous soft exosuit with hip extension assistance for overground walking and jogging"

Figure 1: System actuator and soft exosuit components

State of art: *"Stronger, Smarter, Softer: Next Generation Wearable Robots"* :

- Can be easily align and adjusted.
- Has low inertia.
- Apply force in parallel to the muscle

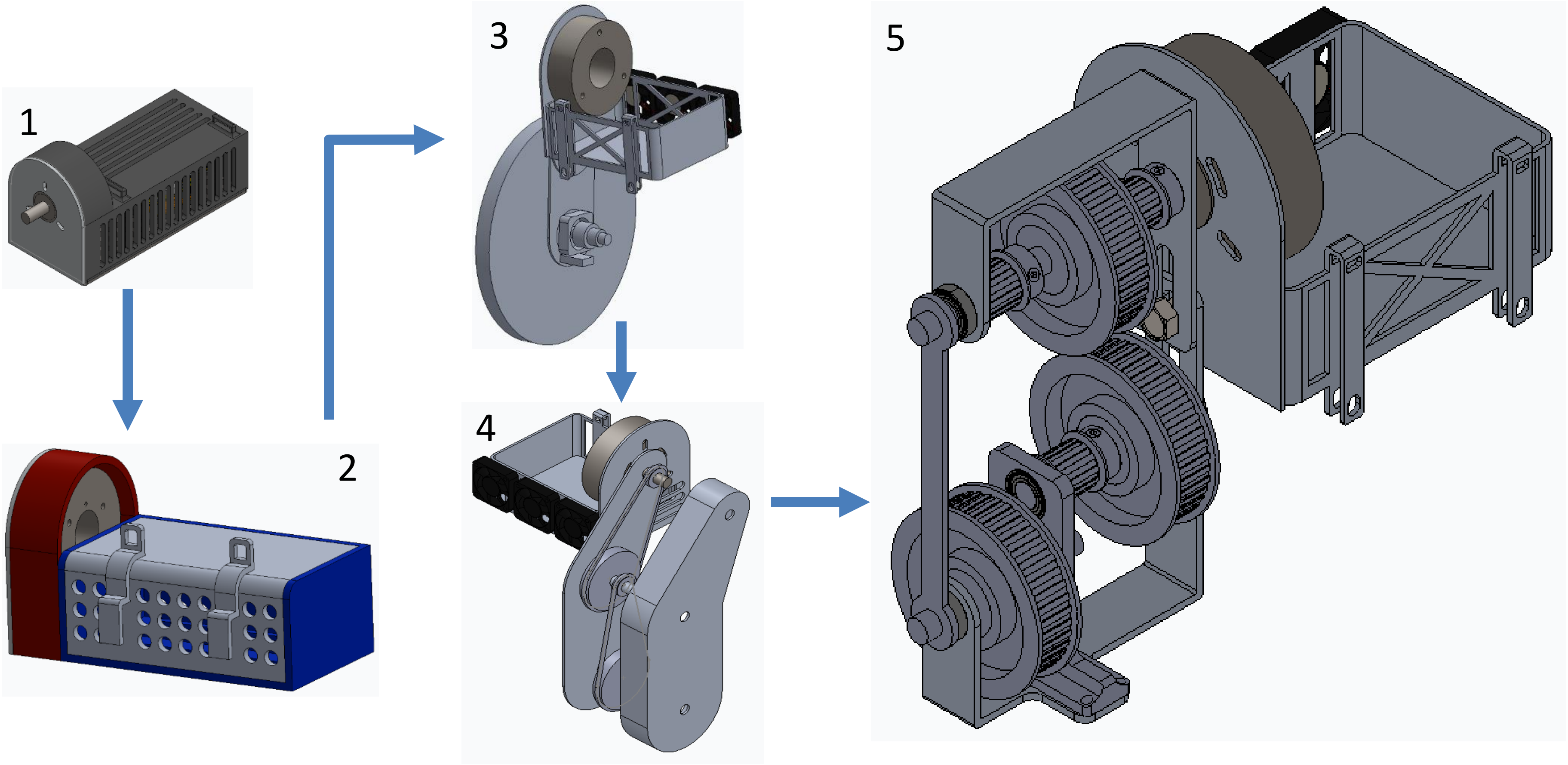




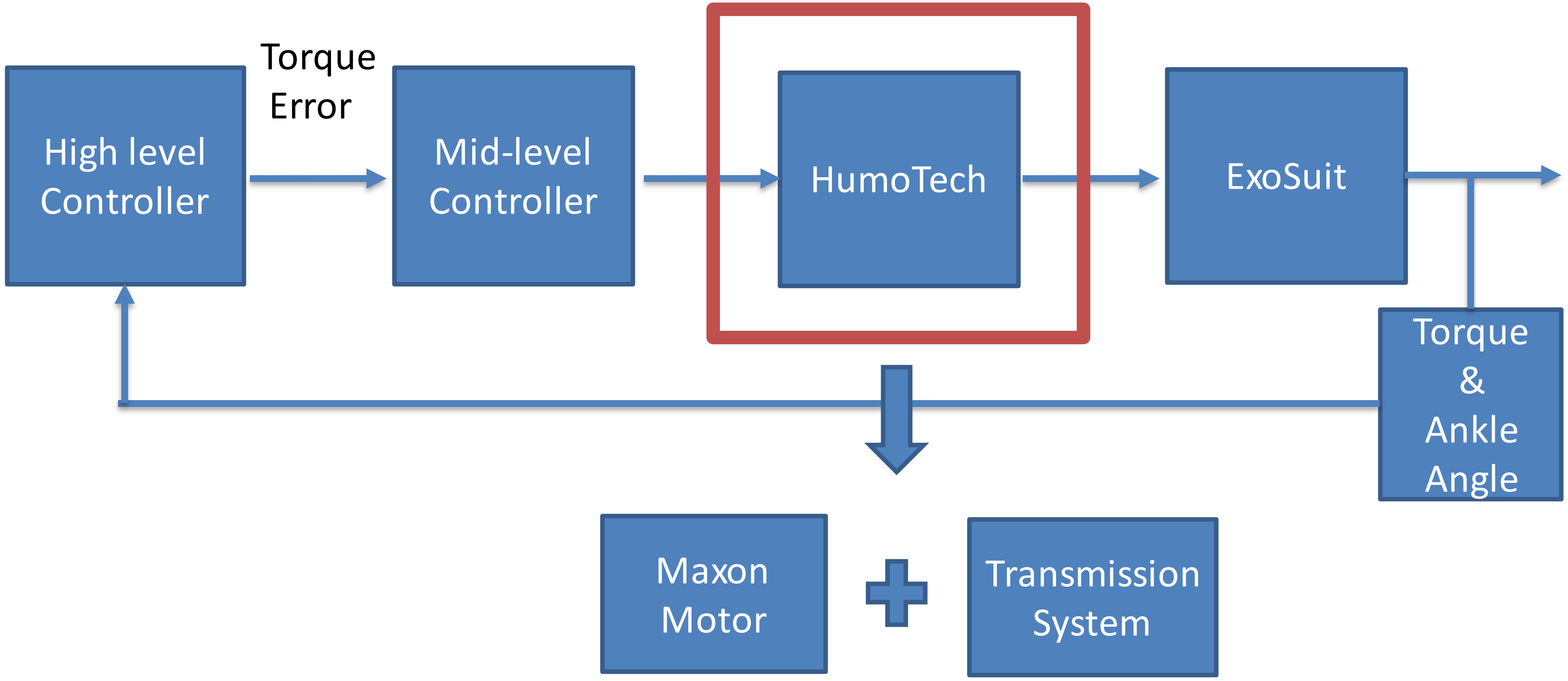
3.2 PRELIMINARY WORK

Hardware Design

- We created a CAD model for Motor Housing, Transmission and its accompanying parts to visualize and iterate the design.



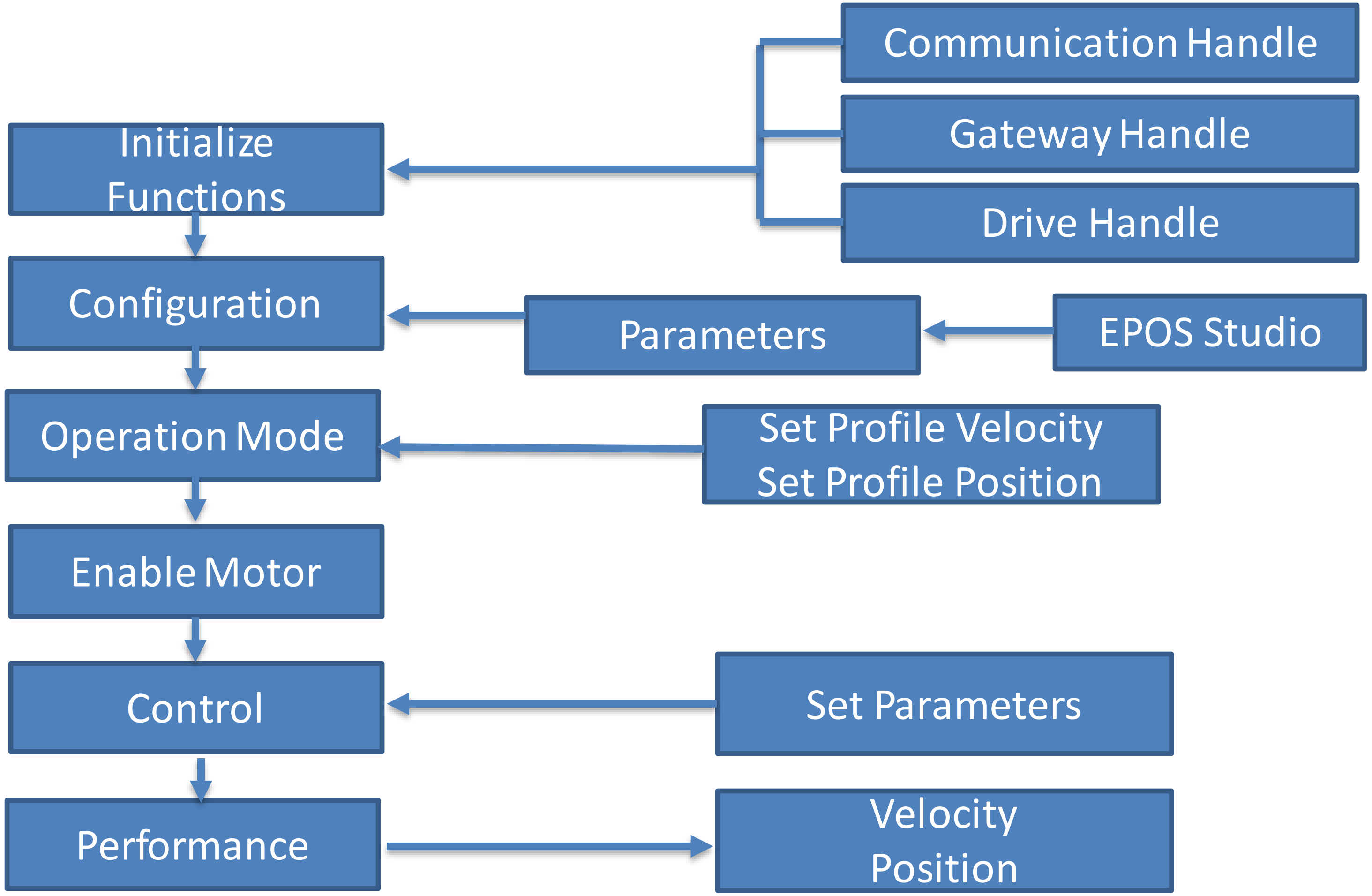
4. RESEARCH



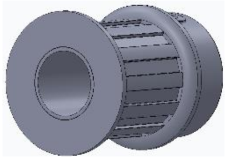
High level Controller
Torque Error
Mid-level Controller
HumoTech
ExoSuit
Torque & Ankle Angle
Maxon Motor
Transmission System

4.1 SOFTWARE

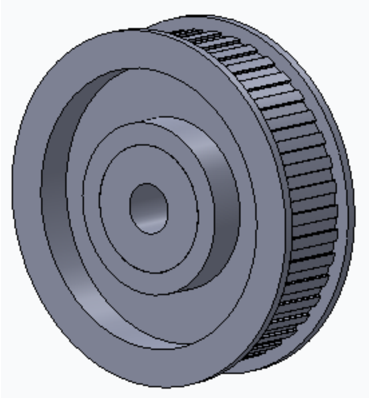
- EPOS2 P Maxon Motor
- Flow of Algorithm



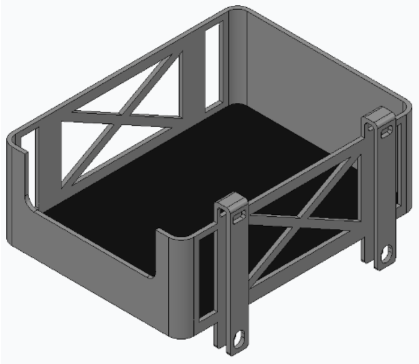
4.2 MOTOR HOUSING AND TRANSMISSION- PARTS



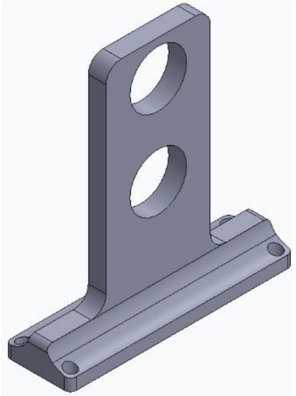
1) 12 Teeth Pulley HTD – Nylon with Aluminum Insert



2) 42 Teeth Pulley HTD – Polycarbonate with Aluminum Insert

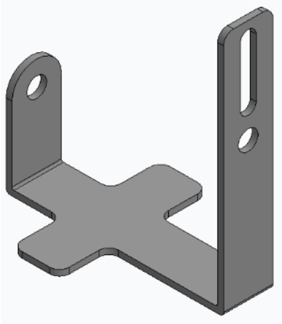


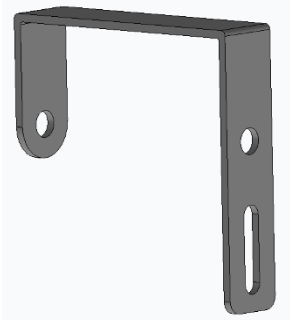
3) Controller Housing - Aluminum



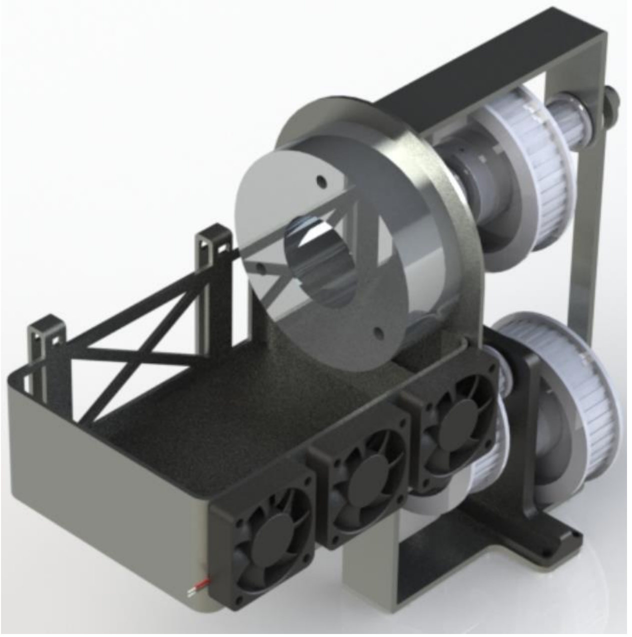
4) Central Pulley Bearing Housing – Thermo Plastic Polymer

5) Motor Mount with Ventilation hole - Aluminum



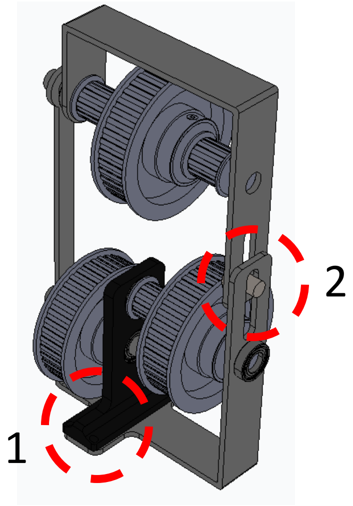


6) Motor Mount with Ventilation hole - Aluminum



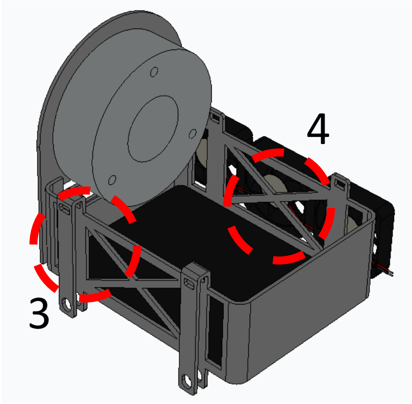
7) Final Assembly

SUB ASSEMBLY & FINAL ASSEMBLY

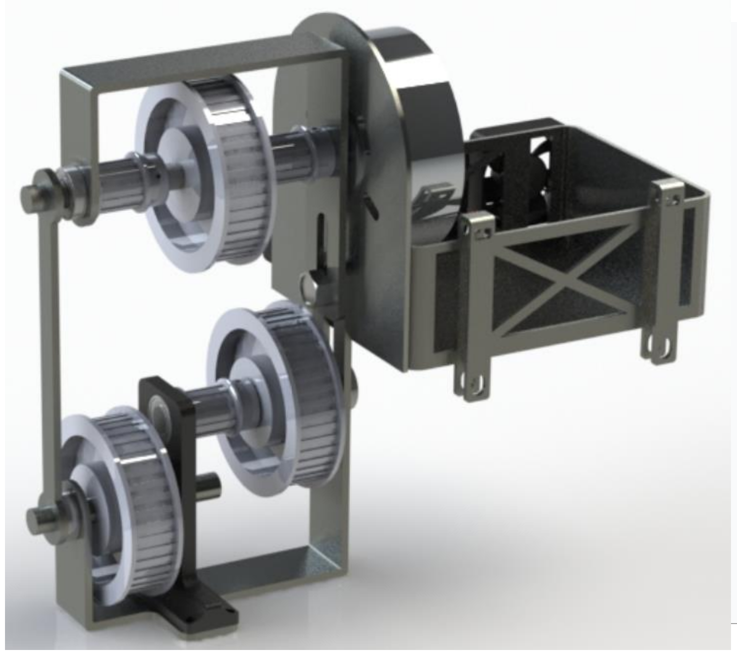


Transmission Sub Assembly

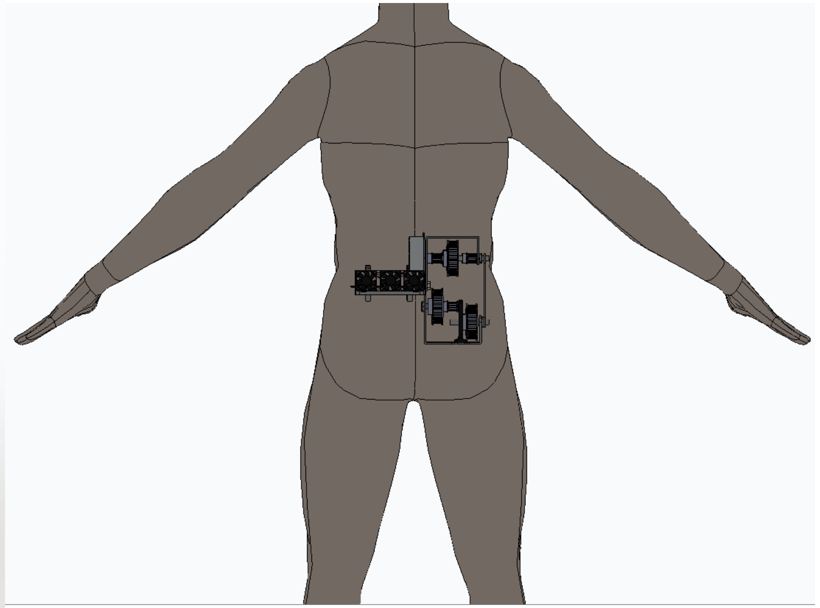
+



Motor Housing Sub Assembly



Final Assembly



Scale Model

Features of Assembly:

1. Lightweight bearing Housing - Custom made Thermoplastic.
2. Adjustable Center Distance for tensioning pulley and changing the ratios if needed.
3. 4 Hooks for mounting to the harness.
4. 40mm Ventilation Fans(Noctua NF-A4x10 FLX Fan)- This fan generates airflow of 4.82 CFM.

Gear Ratio Calculation:

N1 = 12 teeth

N2 = 42 teeth

Ratio = N2*N2*N2/N1*N1*N1

= 42*42*42/12*12*12

= 42.875

5. EVALUATION METHOD

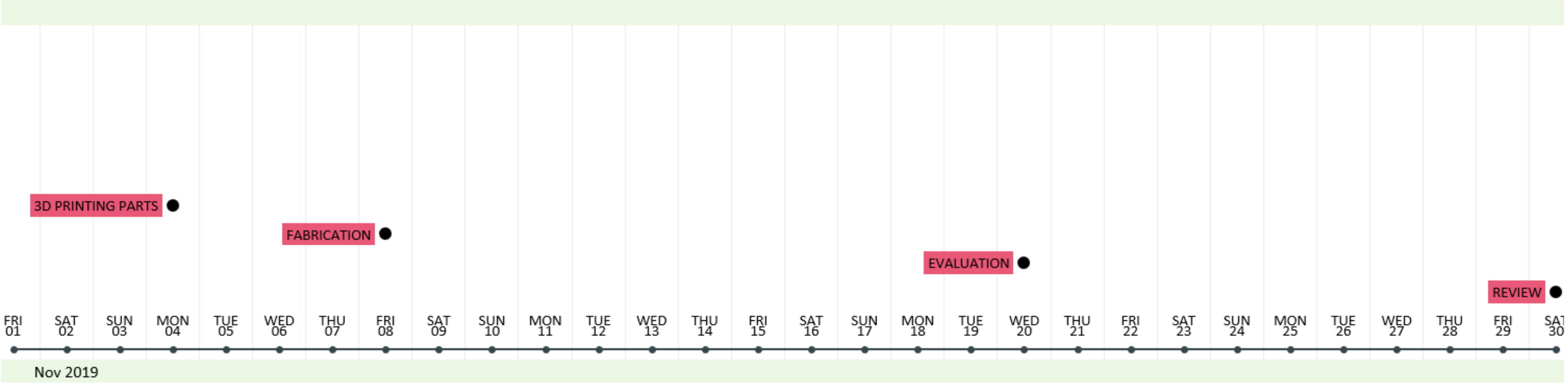
Mobile actuator performance Evaluation:

- Motor Benchtop tests will be carried out.
- After manufacturing of actuator and harness, ankle torque and Ankle angle will be measured. Obtained torque trajectory will be compared with desired torque trajectory.
- Delay for motor actuation will be calculated. Aim is to bring it down to 3-4 ms.

Metabolic costs reduction:

- Metabolic cost will be measured for each iteration of mobile actuator.
- Final design will be selected according to minimum metabolic cost.

6. TIMELINE & PROJECT MEMBER ROLE



ACTIVITY	START	END
Project Start	15-09-19	
LITERATURE SURVEY	15-09-19	26-09-19
ETHERCAT	26-09-19	18-10-19
MOTOR CONTROL	26-09-19	10-10-19
HARNESS DESIGN	27-09-19	08-10-19
MOTOR HOUSING DESIGN	27-09-19	14-10-19
TRANSMISSION DESIGN	27-09-19	14-10-19
HEAT SINK DESIGN	04-10-19	12-10-19
FINAL ASSEMBLY OF 3D MODEL	15-10-19	23-10-19
3D PRINTING PARTS	23-10-19	04-11-19
FABRICATION	26-10-19	08-11-19
EVALUATION	09-11-19	20-11-19
REVIEW	20-11-19	30-11-19
Project End	01-12-19	

Name	Role
Aniket	Software designing and tuning
Abhishek	Designing and fabrication of motor housing and transmission.
Aditya	Designing and fabrication of motor housing and transmission.
Anirudh	Designing and fabricating upper side and motor housing attachment
Harshul	Designing and fabrication of Lower side of exosuit

7. RISK MANAGEMENT

- The major risk factor is near the hip where motor is housed, the vibrations caused by motor could lead to damage or misalignment of the hip, to prevent this the exosuit will have soft padding between the hip and the housing to give comfort.

- There will be high torque produced at the ankle from the motor which makes it important to have robust control algorithm so as not to cause injuries.

- Managing Time to manufacture the housing and transmission is critical for the completion of this project.

8. BROADER IMPACTS

- This soft exoskeleton light weight and low-cost manufacturing makes it easy to wear and easy to manufacture and could be implemented in production basis.

- The exoskeleton could be further use to do more research in improving the efficiency of suit to reduce energy cost.

9. ACKNOWLEDGMENT

- We would like to thank Prof. Kim for the opportunity to work on this project and giving access to her lab and the equipment.

- We would also like to thank An-Chi He for his continued support, Guidance and mentorship without whom progress in this project would not have been possible.